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Cattle Weight Estimation App

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Executive Summary

Western ranches are expansive providing a significant challenge to consistently weigh cattle. Gathering, transporting, and weighing cattle from pastures is resource intensive, and places stress on cattle, leading ranches to rarely weigh their cattle. Cattle weights provide valuable data to ranchers for sales contracts, rangeland health, and other operation decisions. This project aimed to develop a solution to this problem, exploring options which would be cost effective and easily adopted by ranchers.

Throughout the initial research process, two companies were found which are in the process of developing a tool to easily estimate cattle weight. Both companies are utilizing AI to develop a solution and either a phone or 3D camera. Multiple methods were researched for the development of a unique solution including photogrammetry, point-cloud, cameras paired with light detection and ranging sensors, and pixel dimensioning methods. The methods were evaluated for potential accuracy, ease of use, cost, and technical requirements. Different platforms to capture the data were also explored including a smartphone camera, drone, and robotic crawler. Utilizing images taken from a smartphone camera, a pixel dimensioning method, and a regression model was determined to be the optimal method to create and test.

To create a regression model, weight, body length, hook to hook, hip height, and shoulder height were recorded from 41 female Angus cattle. Two sets of data were randomly created, one for the creation of the regression model and a validation data set. A regression model which used body length to predict weight, was selected and used 36 data points. The R^2 value is 0.74 and had p-values of less than 0.05 for the intercept and coefficient. The validation data was used to test the regression, with an average weight prediction error of 5%.

A pixel method was developed to use a known length of the cattle's ear tag to determine the length of one pixel in a picture. The cattle's body length is then determined from a picture taken immediately after they are weighed. This length is then used in the regression to predict the weight. This method was tested with pictures taken of the cattle used to develop the regression model. Utilizing Microsoft paint to find pixel distances, this method had an average error of 12% for estimating body length, and an average error of 26% for body weight. Error in this method is highly dependent on the angle of the cattle with respect to the camera, with perpendicular being the most accurate.

HerdScale, an Android app, was developed to allow ranchers to easily obtain weight measurements utilizing the pixel method, and regression model. Users input breed, date, width of ear tag and the tag identification. Users are then guided through the image capturing process, place markers indicating the tag width and body length in the picture and are then provided with an estimated weight. The app was tested on four market steers and had an average error of 22%.

In future iterations of this tool, it is recommended to implement weight tracking, measurement calibration, data analysis, herd management, model improvement, and multiple breed models.

Introduction

Problem

Accurate weight measurements are beneficial to all cattle operations, and allow ranchers to make data informed decisions, in multiple aspects of their operations. Western ranches are expensive, with cattle often being in pastures miles away from a scale. The time and labor required to gather, and either truck or trail cattle to the scale, is significant and costly. Obtaining weekly weights of cattle, especially yearlings, is beneficial for informing sales contracts, rangeland health, and other operation decisions. The ability to have accurate, easy to obtain weights, would allow ranchers to reduce financial losses. However, due to the cost associated with getting cattle to a scale, obtaining and recording weights, and the stress placed on cattle during this process, cattle are often only weighed a few times a year.

Task

The primary task for the project was to develop an innovative and reliable method for accurately measuring the weight of livestock, specifically cattle, utilizing technology. This comprehensive task involved a multifaced approach beginning with a thorough exploration of current solutions, including an evaluation of various technologies and data analysis tools. The options found were assessed based on criteria's such as accuracy, cost and ease of use, which led to the creation of a project proposal recommending the most effective solution. The project then progressed through the stages of design, development, testing and deployment to ensure the solution met the project requirements.

Overview of weight estimation options

Developing technologies

At the beginning of this project, a search was conducted for existing and developing solutions. A market-ready solution was not discovered; however, two companies are developing the technology required. Both companies utilize AI and machine learning in combination with a 3D or smartphone camera to provide weight estimation.

Frontiers Market

[Frontiers Market](#) is creating a smartphone app which plans to utilize image processing and A.I. algorithms to provide weight estimation, claiming it will be 95% accurate once completed. The app is planned to have the following features: breed id, real-time market value, health scoring, calving prediction, conditioning scores, predictive health intelligence, gain tracking, and lifetime tracking (Frontiers Market, n.d.). The app will require an image of the animal from approximately 10 feet away to provide the estimated weight. Frontiers Market is [searching](#) for ranches to help develop and conduct testing in the future.

Tuul.ai

[Tuul.ai](#), a Berlin-based company, is developing a 3D scanning tool for cattle weight estimation, specifically tailored for Namibia ranchers in South Africa. They claim their tool uses a 3D camera to scan the cattle within three to five meters max and operates offline with data processed on AWS servers. When contact was initiated with this company for possible collaboration, they were still

six months away from having a production-ready solution. They have acknowledged challenges, particularly with accurately scanning black cattle, which further indicates that their tool is still in development. Their solution was not able to meet our deadlines and specific needs for cattle weight estimation over longer distances. Thus, collaboration was not possible at that time.

Self-engineered options

In addition to searching for existing or developing solutions, research was conducted to determine the methods which could be employed to create a tool to perform weight estimation without a scale. In addition, multiple platforms for capturing data were considered. A summary of the findings for methods and platforms which could be utilized is provided below.

Point cloud methods

ZED Camera

ZED cameras are wide-angle 3D cameras which utilize AI to perceive the environment, and range in price from \$400-\$500, depending on the model selected. The cameras require a connection to a computer with a dual-core 2.3 GHz or faster processor, 4 GB of RAM or more, and a Nvidia GPU with computation capabilities greater than 3.0 (StereoLabs, n.d.). The camera must stay connected to a computer when the images are captured to create a point cloud of an object, must be within approximately 3 feet of the object, and the object must stay as still as possible to obtain a highly accurate result. This method is not recommended for use to estimate cattle weight, unless one has significant technological knowledge and time. The most significant challenge posed is keeping cattle still in an open pasture. Even slight movements can significantly impact the accuracy of the point cloud created. A technical method would need to be created to account for cattle movement and may not be accurate. Further technical knowledge would be required to manually or automatically close the holes present in the point cloud where the cattle's hoof meets the ground. Finally, factors such as varying hair length and cattle densities would need to be accounted for.

Multiple camera views with LIDAR

Researchers have utilized multiple cameras in combination with light detection and ranging (LIDAR) sensors to create accurate 3D models of cattle. However, for researchers to create accurate models, the cattle had to be fed in small pens to ensure they remained still. Despite these additional parameters, researchers still often had to remove the shoulder forward from the volume and weight estimates, due to the cattle's movement. In one study, 289 point-clouds were generated, however, only 177 images were usable to generate an accurate volume estimation. Further, manual image processing and truncation of the 3D models was required, requiring regression models to be developed to accurately estimate the cattle's weight (Cozler et al., 2022). This method is impractical for estimating cattle weight on a large scale due to the requirement of multiple cameras, multiple camera views, cattle movement constraints, and manual image processing requirements. Further, even with ideal conditions only approximately 61% of the images were usable, and like other point cloud methods, hair length and varying cattle densities must be accounted for.

Imaging methods

Photogrammetry

Photogrammetry utilizes images captured by a phone camera or drone to create 3D models. The process involves generating point clouds, which can be facilitated utilizing computer tools. However, a significant challenge is the method's current inability to automatically close holes formed at the bottom of the point cloud. Additionally, cattle must remain relatively still for extended periods to ensure accurate data capture, and knowing the density of the cattle is crucial

for precise measurements. Adjustments are also necessary to account for hair length variations, since they can affect the accuracy of volume estimations. Notably, photogrammetry has struggled with correctly estimating barrel volume, indicating areas where further refinement is needed.

Normal images with pixel dimensioning

Normal images with pixel dimensioning can be captured using a smartphone, but this method requires careful setup to ensure accuracy. A known length calibrator must also be included in the image or positioned at a known distance from the cow to provide a scale. The use of a regression equation, such as one developed for Simmental cattle, can help estimate metrics, but new equations are necessary for different types of cattle, including yearlings, cows, bulls, and calves, and other various breeds. To develop these regression models, it is essential to measure and weigh the cattle, and account for changes in weight beyond two years of age. Hair length variations and difficulty in defining the cattle's boundaries in images present additional challenges. Effective use of this method may require training in an AI model or employing techniques like object detection in OpenCV to automatically identify and measure the cattle. Additionally, using an ear tag as a known length reference, ensuring the cow faces the camera, and minimizing movement are critical for achieving accurate results.

Thermal images

Multiple studies have been conducted in which thermal images were utilized to estimate cattle weights. The most accurate weight estimation model utilized thermal imaging to obtain a distinguished outline of Simmental bulls. Utilizing a calibrator, hip and wither heights were found utilizing pixels to determine the respective lengths. Different regression models were used to predict the weight, with the adjusted R^2 values ranging from 0.11 to 0.74 (Stajanko et al., n.d.). Thermal imaging cameras range in prices from \$400 - \$13,400, with some cameras connecting to a smartphone to capture images. This method requires a pixel length calibrator to be heated up, regression models to be utilized, and a consistent image capturing method.

Platforms

Multiple platforms were considered and tested to capture the data required for the weight estimator. Each platform was tested on live cattle in pastures to determine their reactions, as well as feasibility for capturing the required data.

Robotic crawler

A robotic crawler was one of the first platforms tested for this project. A crawler was tested on cattle in a pasture, and it was quickly discovered that the cattle would run whenever the crawler was within 30-50 feet. Sagebrush and grass easily obstructed the camera's view, and the low angle was sub-optimal for capturing images for weight estimation. Battery life is an additional concern, especially for ranches with large pastures.

Drone

Drones offer the advantage of capturing images from multiple angles and can automate data collection at regular intervals. They can cover large areas and fly long distances, making them highly versatile. Drones can be equipped with various imaging technologies, including normal cameras, thermal sensors, and LIDAR, enhancing their ability to gather comprehensive data. Over time, cattle may become accustomed to the presence of drones, potentially reducing stress. However, drones have limitations such as short battery life and a substantial cost, ranging from \$1,500 to \$6,000.

Smartphone

Smartphones offer a convenient option for ranchers who already own them, as they can capture both thermal and normal images. They are highly adaptable and user-friendly, making them an easy choice for many users. However, a potential drawback is the distance to the cattle, which may affect image quality and data accuracy.

After considering all the options available, it was ultimately decided a smartphone app combined with a regression model would be the most viable option to test and develop.

Regression model

Methodology

Weight, body length, hook to hook, hip height, and shoulder height were recorded from 41 cattle. All the cattle were female, with the sample being a mix of Angus cows and yearling heifers. A chute, scale, and measuring stick were utilized to obtain the measurements. Pictures of the cattle were also taken, to be used for testing photo measuring techniques. The measurements were taken as illustrated in the images below.



Figure 1: Picture showing how body length was measured for the regression model data.



Figure 2: Picture showing how hook to hook was measured for the regression model data.

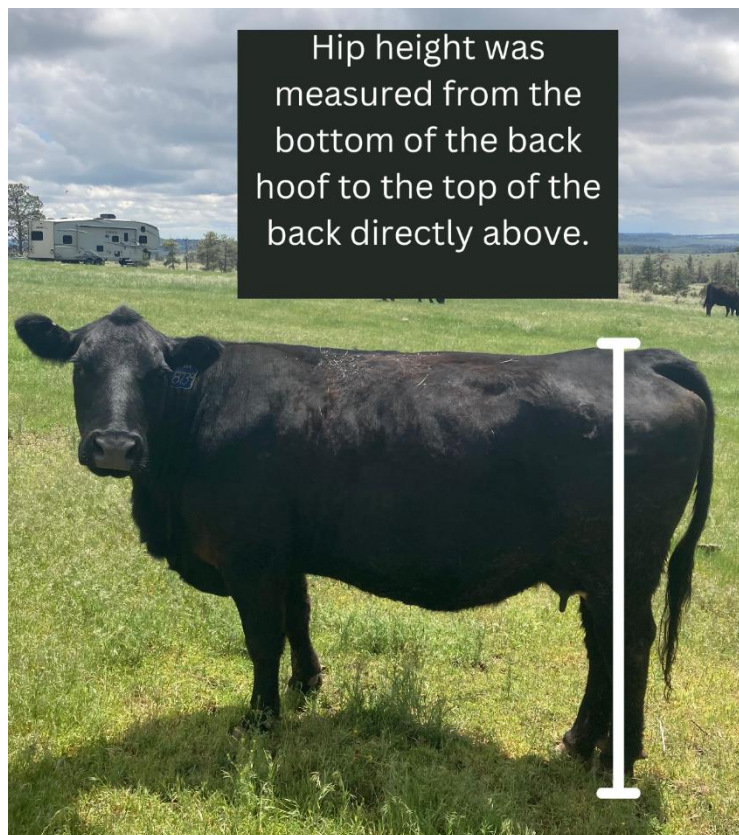


Figure 3: Picture showing how hip height was measured for the regression model data.

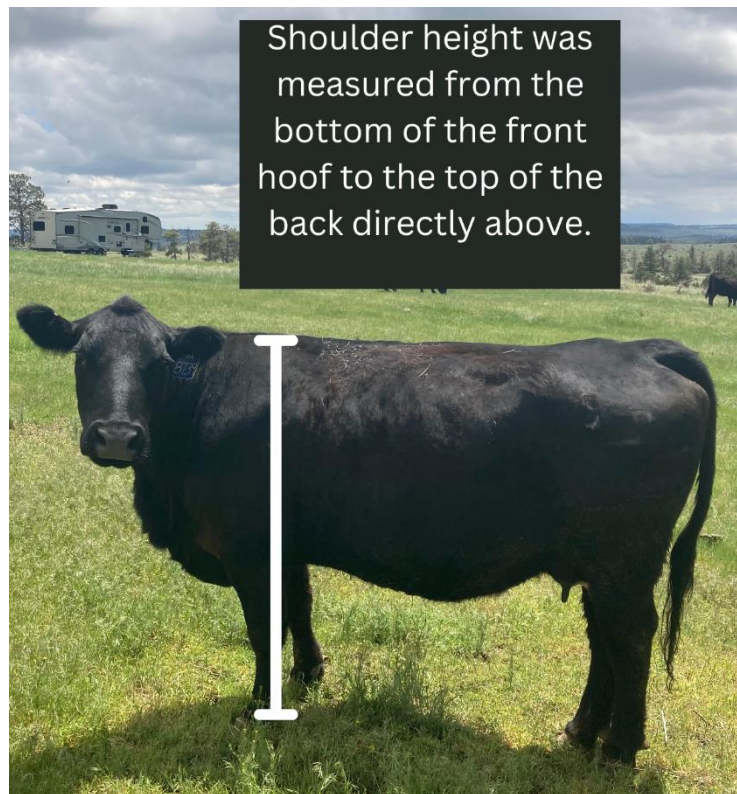


Figure 4: Picture showing how shoulder height was measured for the regression model data.

A random number generator was used to randomly sort the cattle into a set to create regression models, and a validation set. 1-10 was used, with 1-7 corresponding to the set to create the models, and 8-10 to create the validation set. The regression set totaled 36 and the validation set totaled 5. Multiple models were generated, and then checked to see if they met the following assumptions (NCL, n.d.):

- The chosen sample is representative of the population.
- There is a linear relationship between the independent variable(s) and the dependent variable.
- All the variables are normally distributed; to check, plot a histogram of the residuals.
- There are no outliers, (if there are outliers they need to be removed); to check use a test for detecting outliers.
- The independent variables are all linearly independent (no variable depends on the other variables); to check plot the independent values against each other and look for a correlation.
- For multiple regression there should be at least five times as many pairs of data than dependent variables.

The measurements collected were correlated with each other, leading to singular measurement regression models being considered. One model met all the assumptions, which is based on body length, and was used for testing and within the app.

Model

The regression model selected was created using excels regression analytics and has the following regression statistics, ANOVA table, and coefficients.

Regression Statistics	
Multiple R	0.86
R Square	0.74
Standard Error	96.13
Observations	36

Table 1: Regression statistics for the selected regression model

ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	903675.54	903675.5	97.78	1.55E-11
Residual	34	314217.02	9241.68		
Total	35	1217892.56			

Table 2: ANOVA table for the selected regression model

Coefficient Summary Table								
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	-622.38	159.17	-3.91	0.00042	-945.86	-298.90	-945.86	-298.90
Body length (in)	35.92	3.63	9.89	1.6E-11	28.54	43.30	28.54	43.30

Table 3: Coefficient summary table for the selected regression model

The validation data set was then used to test the regression model. Using the measured body lengths of the validation set the regression equation was used to predict the body weight as shown in table 4.

Regression validation				
Tag #	Weight (lbs)	Body length (in)	Predicted weight	Weight error
306	801	40	814	2%
304	743	38	743	0%
8701	1210	52	1245	3%
310	689	39.5	796	16%
			Average error	5%

Table 4: Regression validation data

The regression had an average error of 5% when predicting weight, however a larger validation set would provide more confidence in the accuracy of the regression model.

Pixel measuring

Overview

To predict weight utilizing a picture captured from a smartphone, a pixel measuring method was employed. The method used for the creation of the app requires a side profile picture, an ear tag in view and approximately perpendicular in the photo, and the width of the ear tag. Using the known length of the ear tag, a pixel can be assigned a length. Then utilizing the same measuring methodology used to obtain the lengths of the cattle using a measuring stick, the pixels between the front of the shoulder and the base of the tailhead were determined utilizing Microsoft paint. Utilizing the length of a pixel determined using the ear tag, the body length can be estimated. Similar functionality is included in the app, allowing users to select the front of the shoulder and base of the tailhead, and then the app calculates the pixels between the points, and respective body length.

Testing

To test the pixel measuring method, in conjunction with the regression model, all usable pictures collected the day of weighing the cattle were used for testing. Of the 21 cattle photographed, 16 were usable for testing the pixel method. The estimated body length using the pixel method, and subsequent predicted weight using the regression equation are summarized below in table 5.

Pixel method testing body length				
Tag #	Estimated Body length (in)	Body length error	Predicted Weight (lbs)	Weight error
306	46	16%	1041	30%
304	42	11%	888	19%
8701	63	20%	1626	34%
310	37	6%	710	3%
81	46	6%	1046	11%
63	59	14%	1502	14%
309	43	12%	933	28%
8739	58	14%	1474	15%
M04	60	15%	1518	44%
320	44	10%	941	34%
3A	56	43%	1403	79%
7003	51	1%	1203	1%
3B	47	18%	1069	38%
31	40	1%	817	10%
30	44	1%	942	11%
302	45	5%	992	45%
	Average error	12%		26%

Table 5: Summary of the estimated body lengths utilizing the pixel method, and predicted weight utilizing the regression equation

Both errors were measured as absolute errors to understand the overall magnitude of the estimation errors, regardless of if the estimated measurement or predicted weight was over or under. Utilizing the pixel method to estimate the body length of the cattle has a significant error of 12%. This error likely contributes to the higher predicted weight error, especially when compared to the validation sets error for the regression. The pixel method is prone to error due to any variations in the angle of the ear tag or body of the cattle when compared to the camera taking the picture. It is virtually impossible to get cattle to stand in the perfect position, so this error will always be present when utilizing a pixel method to estimate the lengths of cattle.

Phone measuring apps

Apps claiming to be able to measure objects, including the heights of humans, were tested for accuracy. The Apple measure app was first tested, and it was found it was unable to measure human heights. Tape measure, Tape measure Ruler, Map AR Ruler 3D, and tapemeasureapp were other apps which were tested. While they were able to measure human height, the most accurate app was only within two feet. Due to the large errors found with these apps to measure human heights, they were not further tested on cattle.

App

Key Features

HerdScale is an Android application developed by 9H interns Selma Samet and Rudy Nesvik using Android Studio, designed to provide an intuitive and streamlined user experience for estimating cattle weight. The app guides users through a seamless process that includes data input, image capture, and weight estimation using a regression model.

HerdScale boasts several key features designed to enhance user experience and functionality. Its user-friendly interface is intuitively designed to ensure smooth navigation, making data input and following step-by-step instructions straightforward. The platform integrates advanced image processing capabilities, which, in conjunction with its regression model, deliver accurate weight estimations. By employing sophisticated regression techniques, HerdScale provides precise weight predictions based on captured images and input data. Additionally, the interactive features within the system offer engaging tools that improve data management, making it both more effective and enjoyable. Users also benefit from comprehensive data export options, which facilitate easy saving, sharing, and analysis of their information.

User Interface

The UI design prioritizes simplicity and ease of use, aiming to streamline the weight estimation process for users with minimal technical expertise. This interface includes clear navigation, instructional elements and interactive components to ensure a smooth user experience.

Welcome Page

- The welcome page displays the app's logo and a description to provide users with a brief overview of the app's functionality while setting tone for an engaging experience.

- It includes “ESTIMATE NEW WEIGHT” button to start the first step of cattle weight estimation.

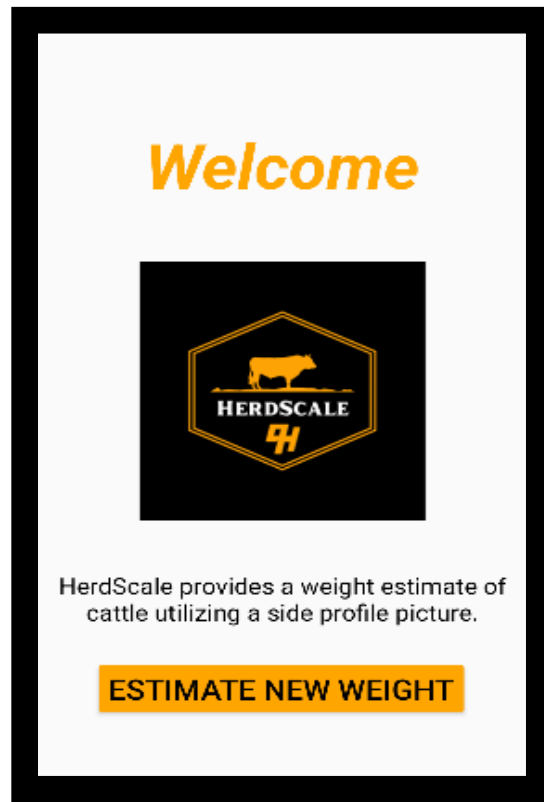


Fig 1: Welcome Page

Screen 1: Input Screen

- Users are prompted to enter details including:
 - ◇ Name: text input
 - ◇ Breed: text input
 - ◇ Date: calendar
 - ◇ Width of ear tag: numeric input in inches
 - ◇ Tag ID: text input
- Navigation: screen includes “Back” and “Continue” button to either go back to welcome page or proceed to the next screen.

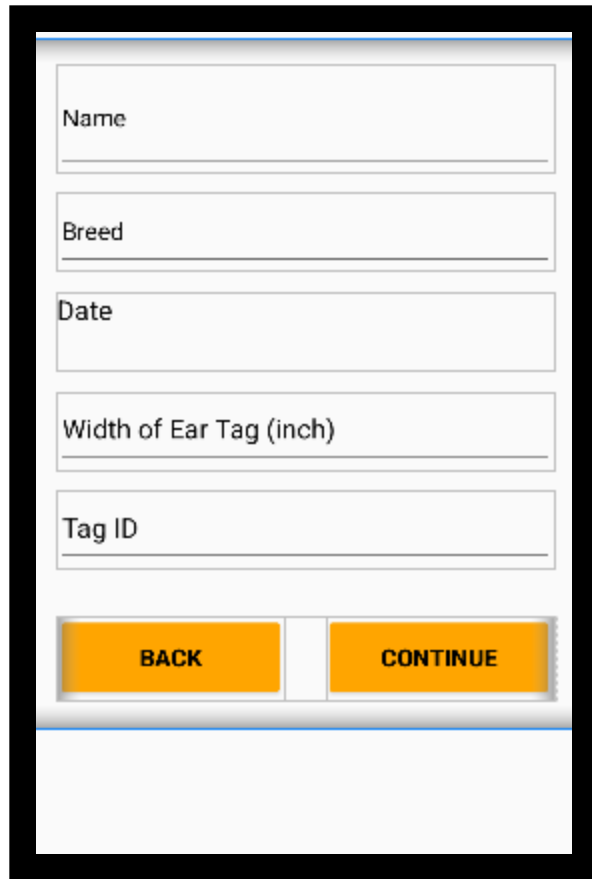


Fig2: Screen where user enters cattle details

Screen 2: Image Capture Screen

- Instructions: user will be able to read clear instructions on how to take a side profile picture of the cattle to ensure clear measurements.
- An Example picture is provided to illustrate correct angle and framing.
- A “Take Side Picture” button allows users to take the picture using the device’s camera along with its zooming features. The user will also be able to retake the picture if needed.
- A “Continue” button is used to proceed to the next step.

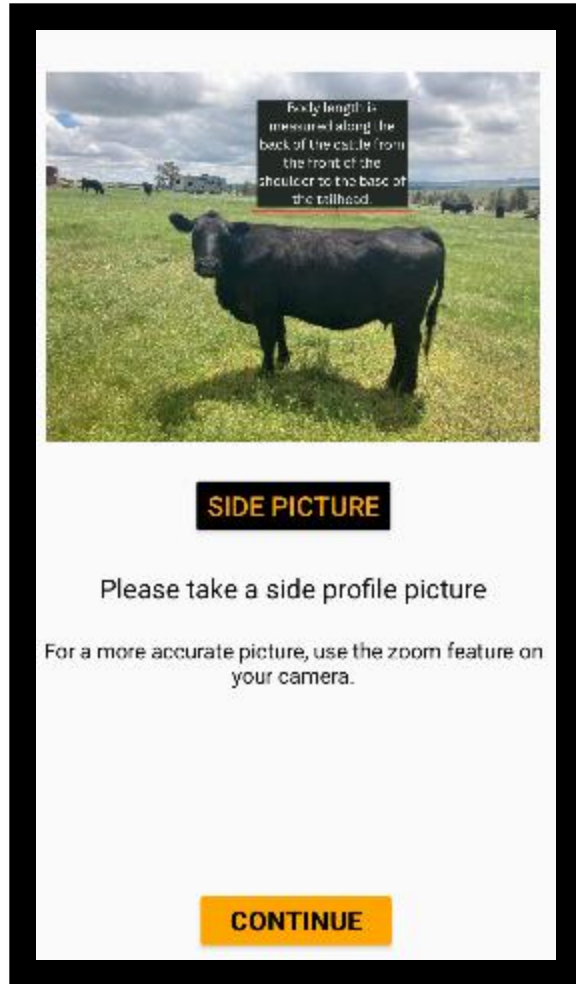


Fig3: Screen where user will take side profile picture of cattle

Screen 3: Measurement Screen

- Instructions and Visual aids are included to help users place markers accurately.
- Point Markers: Users will draw four point markers on the captured image to indicate ear tag length and body length of the cattle.
- Zoom option: Users can zoom in and out of the captured image to allow precise marker placement.

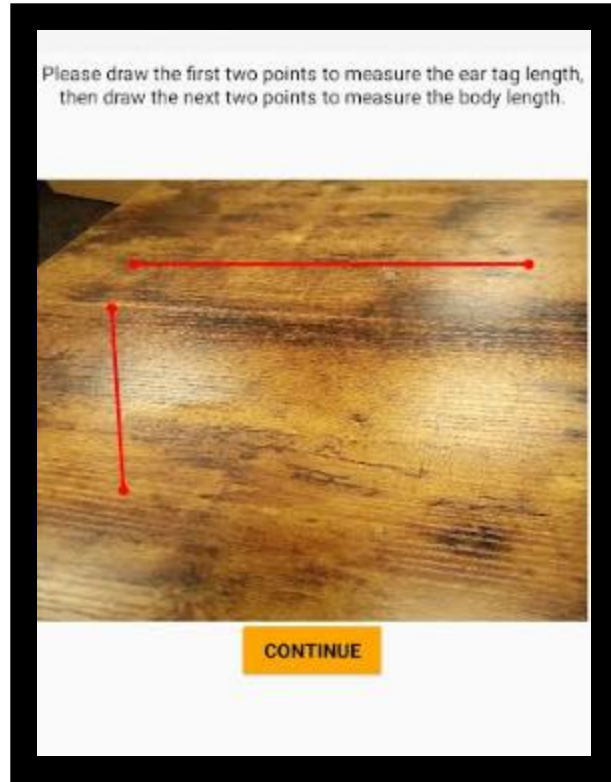


Fig 4: Screen where user will draw point markers to obtain ear tag length of the cattle and its body length.

Screen 4: Result Screen

- Estimation output: The regression model used in the app will process the measurements taken and display the estimated weight of the cattle.
- Rancher Guess feature: Users can input their own weight guess and obtain a comparison with the app's estimate weight.
- Export button: This is an option that users can use to export all the data used during the weight estimation process in a CSV format for further analysis. This includes user inputs, measurement results and estimated weights to allow users to save information for later use.

Estimated weight for cow is: lbs
941 lbs

857

SUBMIT GUESS

EXPORT DATA

Your guess: 857 lbs
Difference: 84 lbs
Percentage off: 8.93%

BACK TO HOME

Fig 5: Screen where user will obtain estimated cattle weight and export all related data into CSV format.

A	B	C	D	E	F	G	H	I
Name	Breed	Date	Width of E Tag ID	Ear Tag Le	Body Leng	Rancher's	Estima	
test1	cow	13-Aug	2.5	1000	152.0763	7.990846	100	-3

Fig 6: Result exported CSV file that contains user data, image measurements, rancher guess and weight estimates.

Backend

Data Handling

Data handling in the app begins with user input on Screen 1, where entered data is stored and processed for later use in the weight estimation equation. Additionally, the app captures a side profile image, which is then stored externally on the user's phone. To extract the necessary measurements from the photo, the app utilizes advanced image processing algorithms. This combination of data entry and image analysis ensures accurate and efficient weight estimation.

Measurements and Calibration

The app employs algorithms to convert drawn point markers into measurable data, calculating the lengths necessary for weight estimation. This data is then processed using a regression model to estimate cattle weight based on the measurements taken. The model's coefficients, which are stored within the app, are utilized to make precise weight predictions.

Export functionality

The app can generate a CSV file that includes user inputs, image data, and weight estimates. This file format allows users to efficiently store and analyze their data.

Error handling

The app features a mechanism to manage input errors, such as invalid text formats or incomplete fields, by providing prompts for correction. Additionally, measures have been implemented to ensure the app operates efficiently, including optimizing image processing and regression calculations to deliver timely results.

Rancher guess feature

The Rancher Guess feature in the HerdScale app allows users to input their own estimated weight for cattle and compare it with the app's weight estimate. This approach offers a straightforward way for users to gauge the accuracy of their guesses, thus enhancing the app's utility.

App Testing

The app underwent two rounds of testing to evaluate its performance and accuracy. The first round yielded unsatisfactory results. Four cows were tested with an ear tag width of 2.5 inches, but the results were negative and nonsensical, indicating issues with the initial calibration due to not being able to zoom in on the pictures taken and draw precise point markers.

In the second round of testing, the zoom option was utilized to enhance image quality and measurement accuracy. This adjustment led to improved results, as shown in the table below:

Second round of app testing				
Tag ID	Estimated Weight (lbs)	Actual weight (lbs)	Width of ear tag (inch)	Percentage Error (%)
Yellow	1073	1286	2.5	16.6
Pink	996	1228	2.5	18.9
Blue	780	1402	2.5	44.3
Red	980	1066	2.5	8.1
Average Percentage Error 22%				

Table 1: Summary of results from the second round of cattle testing

These results demonstrate the app's progress from the initial testing phase to improved performance with adjustments. Further refinements may be necessary to enhance accuracy and reliability.

Recommendations

To further enhance HerdScale's functionality and user experience, the following improvements are recommended:

Weight Tracking

Introducing functionality to track and visualize weight changes over time will allow users to monitor growth of cattle and make informed decisions.

Enhanced Measurement Calibration

Implementing automatic calibration tools and providing a calibration guide will help improve the accuracy of measurements by adjusting for image distortions and varying angles.

Advanced Data Analysis

Adding predictive analytics to forecast future weight trends and comparative analysis features will help assess individual and herd-wide performance.

User Profile and Herd Management

Developing user profiles to manage multiple cattle entries and providing a dashboard for an overview of herd weight and growth data will improve overall herd management.

Model Accuracy Improvement

Refining the regression model by incorporating more diverse data sets and conducting additional testing will enhance its accuracy. This includes, for example, expanding data collection beyond the current breed type to improve predictions for a broader range of cattle breeds.

Appendix

Regression model data

Data for regression					
Tag #	Weight (lbs)	Hip height(cm)	Shoulder height(cm)	Hook to hook (in)	Body length (in)
311	687	142	134	14	34
81	1180	158	155	17.5	44
63	1315	164	161	22	52
309	729	148	138	14.5	38.5
305	649	141	135	12	37
8739	1280	162	155.5	23	51
M04	1055	158	153	17.5	52
0409	1190	158.5	150	20.5	50
301	719	145.5	138	15	38.5
320	701	142.5	136	15	39.5
071	1175	160	157	21	48
3A	783	144	137	15	39.5
7003	1215	158	155	21	50.5
3B	775	145.5	139	14	40
31	745	141	140.5	15	39.5
30	849	148	139.5	19	44
302	683	146	140.5	14.5	43
9H	1082	141	147	18	46
9H	778	134	141	16	39
9H	976	144	149	16.5	39.5
9H	902	134	143	16	41
9H	1098	146	152	17	47.5
9H	882	139.5	145.5	16	45
9H	1070	141	147	17	44
9H	1020	133	143	15.5	43
9H	858	135	144	17.5	39.5
9H	936	148	158	16.5	43.5
9H	1068	145	144	17	46.5
9H	1114	144.5	147	15.5	47
9H	974	142.5	145	17	44.5
9H	780	136	147.5	16	39.5
9H	924	144	153.5	17	46
9H	1102	140.5	147	16.5	45
9H	986	143	147	16	46
9H	786	134	140	15.5	41
9H	904	143	145	16	45

Table A.1: Measurements taken for the regression model

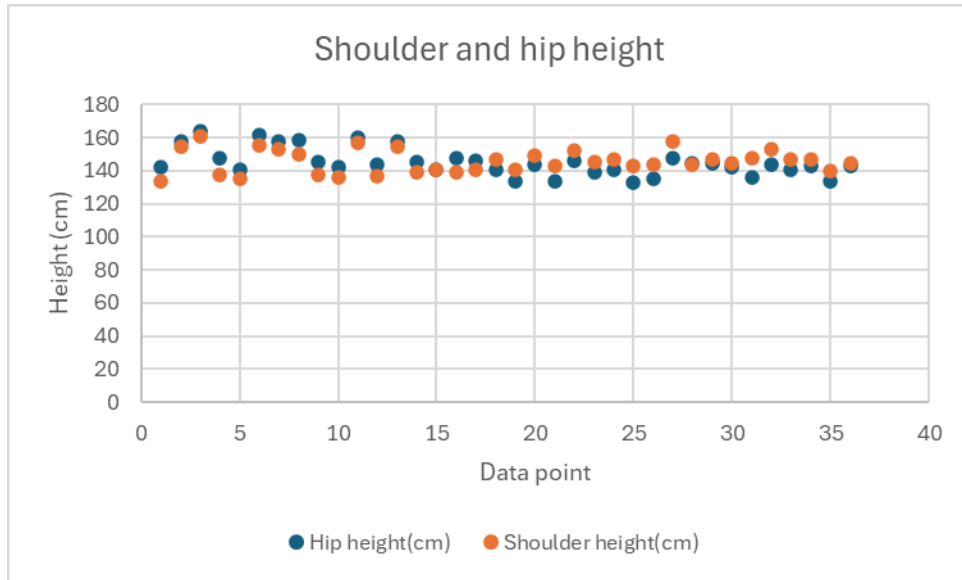


Figure A.1: Plot used to check for outlier data in the shoulder and hip height measurements

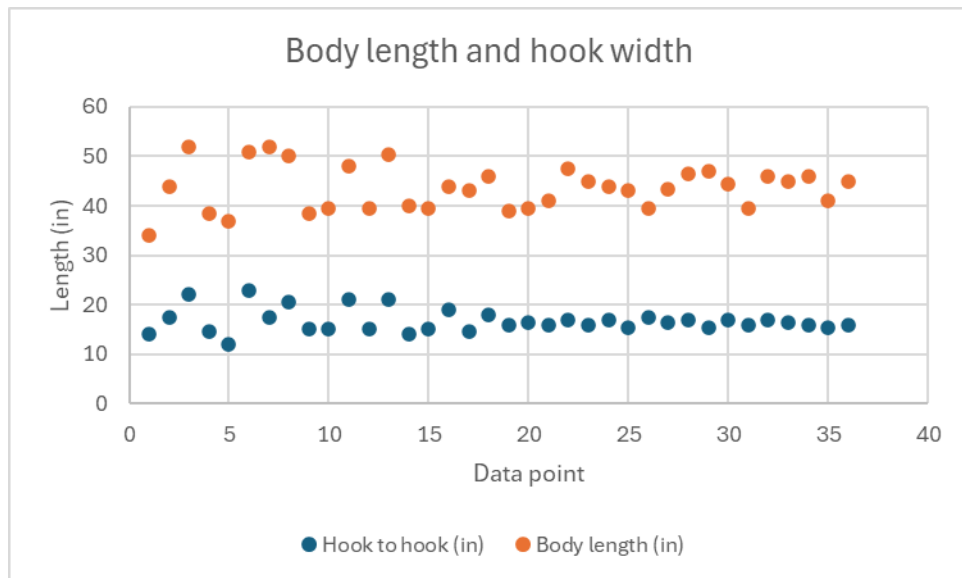


Figure A.2: Plot used to check for outlier data in the body length and hook width measurements

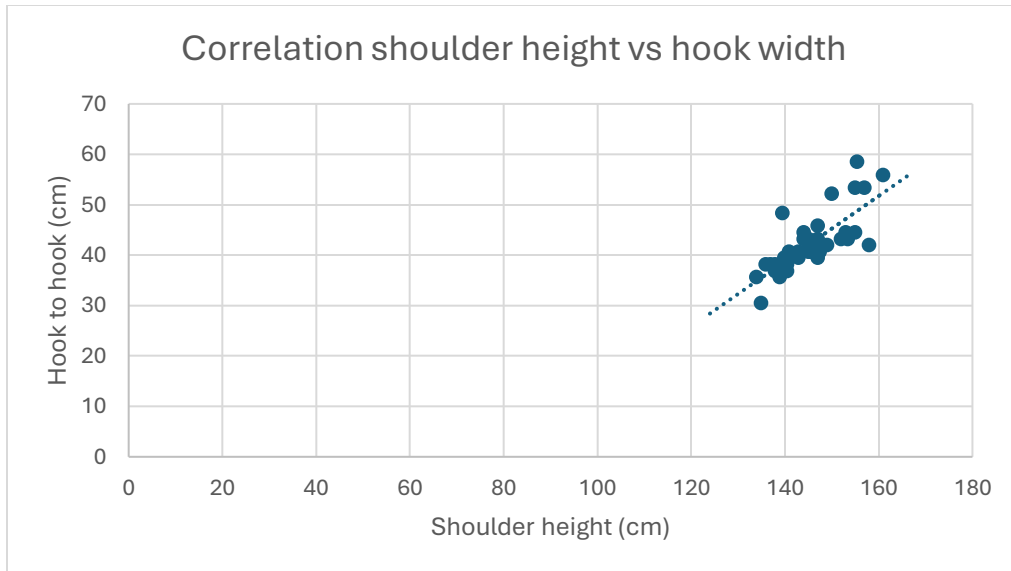


Figure A.3: Correlation between shoulder height and hook width measurements

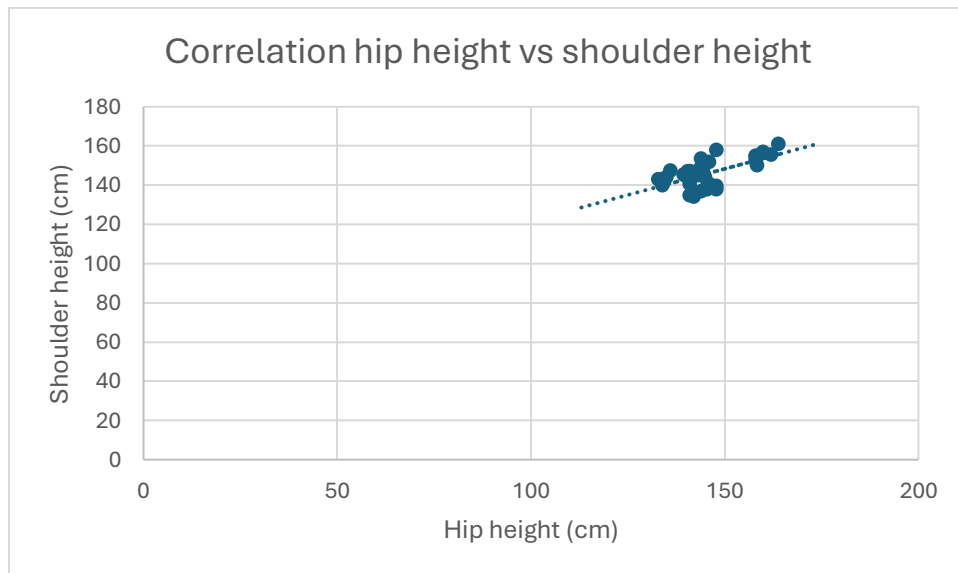


Figure A.4: Correlation between hip height and shoulder height measurements

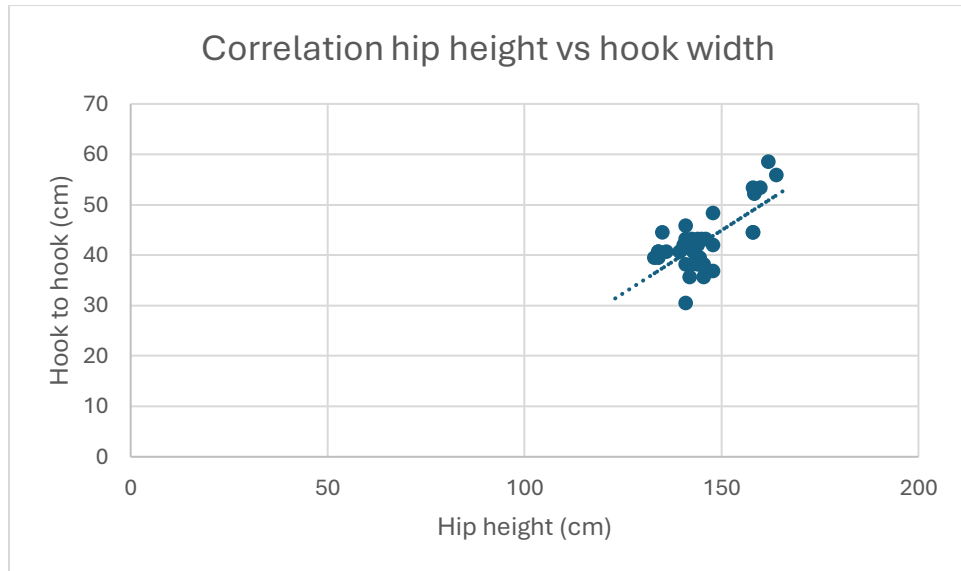


Figure A.5: Correlation between hip height and hook width measurements

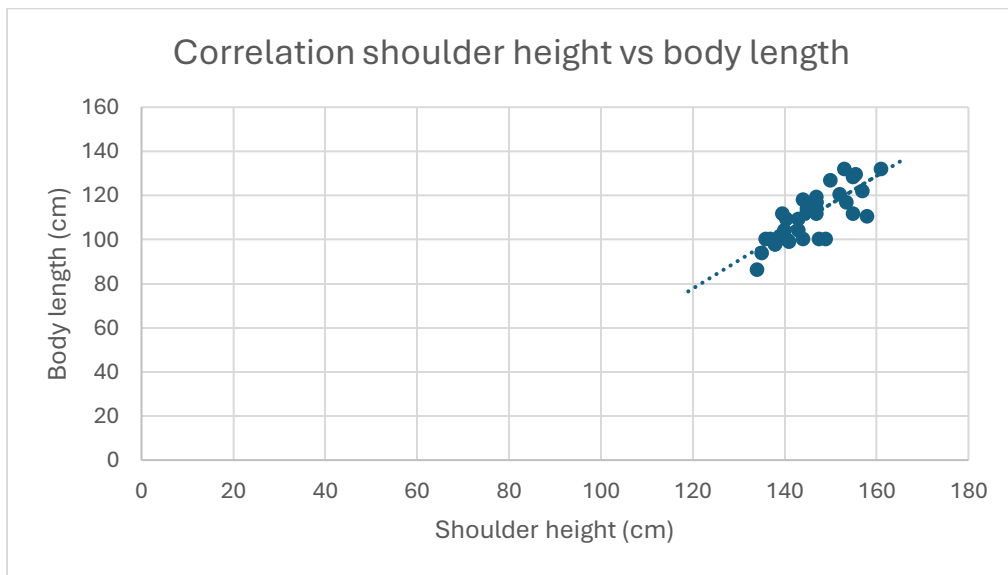


Figure A.6: Correlation between shoulder height and body length measurements

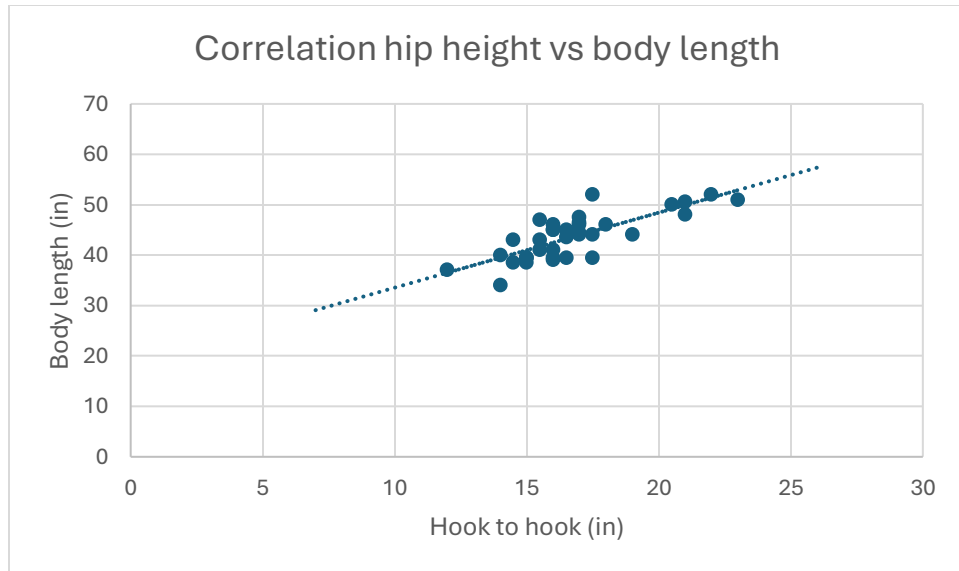


Figure A.7: Correlation between hip height and body length measurements

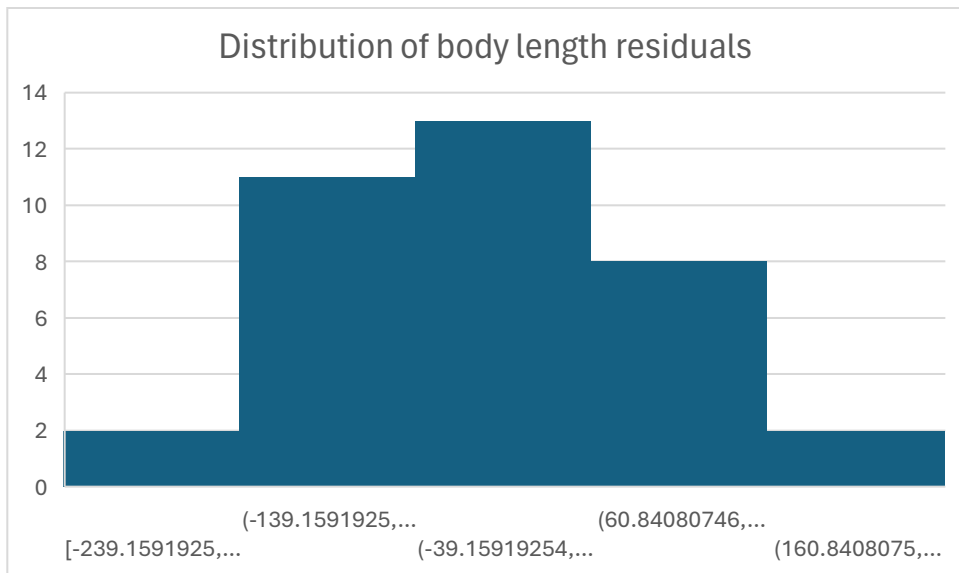


Figure A.8: Distribution of the residuals of the body length regression model.

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